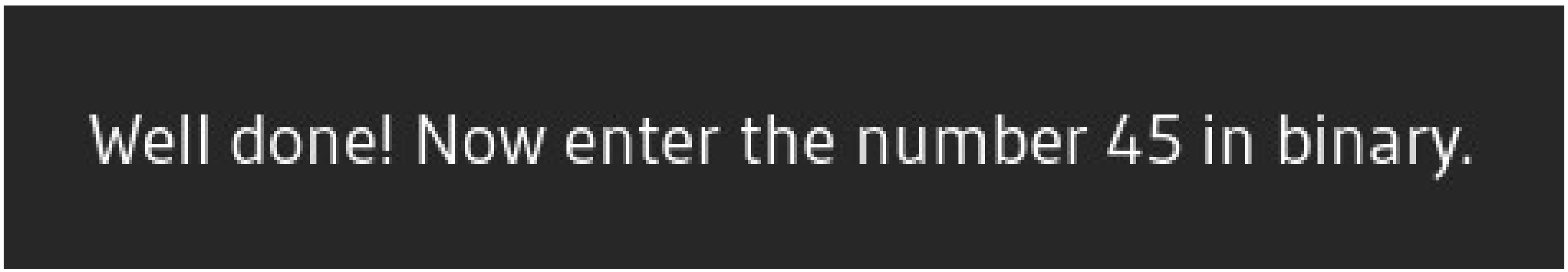
Hack The Web – Binary II



In this challenge, we're told to input the binary number for the decimal number 45 as the answer for the challenge

HTW – Binary II
Binary Numbers

Decimal Number Symbols

0 1 2 3 4 5 6 7 8 9

The regular numbers we use, called decimal numbers, use a base-10 system, where there are ten symbols used for a set

HTW – Binary II
Binary Numbers

```
Binary Number Symbols

0 1 ← each set is two
```

Binary numbers, on the other hand, are another system of counting, where there are only two symbols in a set, hence “binary”, one or two

HTW – Binary II
Binary Numbers

```
0 1 10 11 100 101 110 ⟵ binary

0 1 2  3  4   5   6   ⟵ decimal
```

If we compare counting in decimal versus binary counting, we see that the numbers become very long very quickly

HTW – Binary II
Context - Binary Numbers

```
Computer Machine Code (x86 Chipset)

10010000 ⟵ No Operation
```

The reason why binary numbers are important in computing is because at the very lowest level, computer processors only take in data in the form of zeros and ones, that is binary numbers

HTW – Binary II
Context - Binary Numbers

```
10010000 ⟵ Binary

90       ⟵ Hexadecimal

144      ⟵ Decimal
```

Of course, these binary numbers are usually converted to different number systems, decimal or hexadecimal, when people interact with them, because long strings of binary numbers are not very practical to work with